STRUCTURE OF THE NORMALIZATION OF A BASE IN A NORMAL SOURCE WITH LOCALLY FINITELY MANY COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-normalization-normal is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 14503-14525 and proof span 14527-14585 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof ends with the author's sentence 'We omit the verification that it is unique.' for the uniqueness part; that sentence is kept verbatim, and the construction of the canonical factorization is carried out in the counted span.

Editorial note. This card was previously skipped because the shared gate substring-matched my chapter name against the published items of more-morphisms.tex; after the Lead repaired the gate the candidate was re-gated (RESULT=OK) and opened.

Editorial note. 8 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The structure statement for the normalization in a normal source, established earlier in the same chapter (and established by a previous card of this lane). Cited at proof lines 14527-14531 and 14537-14539. Gives that the normalization is a disjoint union of normal integral schemes and that the intermediate morphism induces a bijection on irreducible components.
- The characterization of the normalization morphism as integral, and the compatibility of normalization with passing to the reduction, both cited from the same chapter. Cited at proof lines 14536-14537 and 14565-14566. Give integrality of the normalization and let the uniqueness statement be reduced to the reduced case.
- An integral morphism is universally closed, and the description of the normalization by integral closures and by the construction through a dominant morphism from a normal source, cited from the same chapter. Cited at proof lines 14542-14544, 14553-14555 and 14570-14570. Turn dominance

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into surjectivity, give the factorization statement, and supply the local algebra used for the uniqueness statement.
- A morphism to a reduced scheme factors through the reduction, and a normal ring equals its integral closure in its total ring of fractions. Cited at proof lines 14566-14568 and 14580-14582. Reduce to a reduced base and identify the target of the integral closure map with the normal coordinate ring.

Source reference. Original lemma 034M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4097–4097. Same-chapter reference used in the retained author text at source line(s) 14579. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original equation 035M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4133–4133. Same-chapter reference used in the retained author text at source line(s) 14387, 14406, 14413, 14462, 14528, 14551, 14996. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 035I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4129–4129. Same-chapter reference used in the retained author text at source line(s) 14027, 14033, 14059, 14409, 14538, 14556, 14913, 15004. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 035P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4136–4136. Same-chapter reference used in the retained author text at source line(s) 14491, 14535, 14570, 14599, 14678, 14700, 15083. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2487–2487. Same-chapter reference used in the retained author text at source line(s) 10976, 11172, 11375, 11380, 11512, 11552, 11624, 11672, 11769, 14152, 14165, 14544. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 035L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4132–4132. Same-chapter reference used in the retained author text at source line(s) 14532, 14539. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 035O: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4135–4135. Same-chapter reference used in the retained author text at source line(s) 14534, 14561, 14597. It is supplied only as a

navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0356: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4117–4117. Same-chapter reference used in the retained author text at source line(s) 291, 987, 10250, 13640, 14408, 14563. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a scheme such that every quasi-compact open has finitely many irreducible components.*

1. *The normalization X^ν is a disjoint union of integral normal schemes.*
2. *The morphism $\nu : X^\nu \to X$ is integral, surjective, and induces a bijection on irreducible components.*
3. *For any integral morphism $\alpha : X' \to X$ such that for $U \subset X$ quasi-compact open the inverse image $\alpha^{-1}(U)$ has finitely many irreducible components and $\alpha|_{\alpha^{-1}(U)} : \alpha^{-1}(U) \to U$ is birational[1] there exists a factorization $X^\nu \to X' \to X$ and $X^\nu \to X'$ is the normalization of X'.*
4. *For any morphism $Z \to X$ with Z a normal scheme such that each irreducible component of Z dominates an irreducible component of X there exists a unique factorization $Z \to X^\nu \to X$.*

Proof. Let $f : Y \to X$ be as in (035M). The scheme X^ν is a disjoint union of normal integral schemes because Y is normal and every affine open of Y has finitely many irreducible components, see Lemma 035L. This proves (1). Alternatively one can deduce (1) from Lemmas 035O and 035P.

The morphism ν is integral by Lemma 035I. By Lemma 035L the morphism $Y \to X^\nu$ induces a bijection on irreducible components, and by construction of Y this implies that $X^\nu \to X$ induces a bijection on irreducible components. By construction $f : Y \to X$ is dominant, hence also ν is dominant. Since an integral morphism is closed (Lemma 01WM) this implies that ν is surjective. This proves (2).

Suppose that $\alpha : X' \to X$ is as in (3). It is clear that X' satisfies the assumptions under which the normalization is defined. Let $f' : Y' \to X'$ be the morphism (035M) constructed starting with X'. As α is locally birational it is clear that $Y' = Y$ and $f = \alpha \circ f'$. Hence the factorization $X^\nu \to X' \to X$ exists and $X^\nu \to X'$ is the normalization of X' by Lemma 035I. This proves (3).

Let $g : Z \to X$ be a morphism whose domain is a normal scheme and such that every irreducible component dominates an irreducible component of X. By Lemma 035O we have $X^\nu = X^\nu_{red}$ and by Schemes, Lemma 0356 $Z \to X$ factors through X_{red}. Hence we may replace X by X_{red} and assume X is reduced. Moreover, as the factorization is unique it suffices to construct it locally on Z. Let $W \subset Z$ and $U \subset X$ be affine opens such that $g(W) \subset U$. Write $U = \mathrm{Spec}(A)$ and $W = \mathrm{Spec}(B)$, with $g|_W$ given by $\varphi : A \to B$. We will use the results of Lemma 035P freely. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_t$ be the minimal primes of A. As Z is normal, we see that B is a normal ring, in particular reduced. Moreover, by assumption any minimal prime $\mathfrak{q} \subset B$ we have that $\varphi^{-1}(\mathfrak{q})$ is a minimal prime of A. Hence if $x \in A$ is a nonzerodivisor,

[1] This awkward formulation is necessary as we've only defined what it means for a morphism to be birational if the source and target have finitely many irreducible components. It suffices if $X'_{red} \to X_{red}$ satisfies the condition.

i.e., $x \notin \bigcup \mathfrak{p}_i$, then $\varphi(x)$ is a nonzerodivisor in B. Thus we obtain a canonical ring map $Q(A) \to Q(B)$. As B is normal it is equal to its integral closure in $Q(B)$ (see Algebra, Lemma 034M). Hence we see that the integral closure $A' \subset Q(A)$ of A maps into B via the canonical map $Q(A) \to Q(B)$. Since $\nu^{-1}(U) = \mathrm{Spec}(A')$ this gives the canonical factorization $W \to \nu^{-1}(U) \to U$ of $\nu|_W$. We omit the verification that it is unique. $\square$